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Circulation: Cardiovascular Quality and Outcomes

ORIGINAL ARTICLE

Discrimination Experiences and All-Cause and Cardiovascular Mortality: Multi-Ethnic Study of Atherosclerosis

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BACKGROUND: Epidemiologic studies have documented the associations between experiences of discrimination and adverse health outcomes. However, the relationship between discrimination and mortality, and the factors that may moderate this relationship are not well understood. This study examined whether lifetime and everyday discrimination were associated with all-cause and cardiovascular mortality and whether these associations differed by race and ethnicity, gender, and racial and ethnic residential segregation.

METHODS: The study included 1633 Black, 1403 Hispanic/Latino, and 2473 White participants aged 45 to 84 years from the Multi-Ethnic Study of Atherosclerosis, enrolled from 2000 to 2002 and followed across 5 exams (2002–2018). Discrimination was measured using the lifetime discrimination (major experiences of unfair treatment) and everyday discrimination (day-to-day experiences of unfair treatment) scales. Racial and ethnic residential segregation was measured using the *Gi** statistic. Cox proportional hazards regression was used to estimate hazard ratios (HRs) and 95% CIs, adjusting for sociodemographic characteristics, health behaviors, and clinical risk factors.

RESULTS: Each increase in reports of lifetime discrimination was associated with increased all-cause (HR, 1.06 [95% CI, 1.00–1.11]) and cardiovascular (HR, 1.15 [95% CI, 1.04–1.27]) mortality, adjusting for sociodemographic factors, health behaviors, and clinical risk factors. Associations between lifetime discrimination and cardiovascular mortality were observed across all racial and ethnic groups but were strongest and only statistically significant among Black participants (HR, 1.18 [95% CI, 1.02–1.37]). Additionally, in the fully adjusted model, each increase in reports of everyday discrimination was strongly associated with increased cardiovascular mortality (HR, 1.21 [95% CI, 1.03–1.43]). Associations for lifetime and everyday discrimination with all-cause and cardiovascular mortality were not modified by race and ethnicity, gender, or racial and ethnic residential segregation.

CONCLUSIONS: These findings suggest that experiences of discrimination are associated with increased all-cause and cardiovascular mortality.

Key Words: ethnicity ■ heart disease ■ mortality ■ racism ■ residential segregation

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In the United States, racial and ethnic disparities in life expectancy remain a major public health issue. In 2020, the life expectancies of Black men and women were ≈5 to 7 years shorter than their White and Hispanic counterparts.[1] Most studies examining factors contributing to racial and ethnic disparities in mortality have focused on traditional individual-level factors including behaviors, biomedical risk factors, and barriers to medical care. There have been numerous calls to investigate the role of broader upstream factors that structure and perpetuate

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Supplemental Material is available at https://www.ahajournals.org/doi/suppl/10.1161/CIRCOUTCOMES.122.009697.
For Sources of Funding and Disclosures, see page 325.

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WHAT IS KNOWN

- Experiences of discrimination have been linked to adverse health behaviors and cardiovascular disease, especially among racial and ethnic minoritized populations.
- However, little is known about the relationships between accumulated experiences of discrimination and all-cause and cardiovascular mortality.

WHAT THE STUDY ADDS

- In this large multiethnic cohort with nearly 2 decades of follow-up, experiences of discrimination were linked to higher all-cause and cardiovascular mortality with stronger associations observed for cardiovascular than for all-cause mortality.
- Experiences of lifetime discrimination were associated with all-cause and cardiovascular mortality among Black participants.

Nonstandard Abbreviations and Acronyms

HR	hazard ratio
MESA	Multi-Ethnic Study of Atherosclerosis

racial differences in exposures and opportunities affecting health. One such factor is racism, which is manifested in several ways including at the policy, institutional, and interpersonal levels.[2–4]

Several epidemiologic studies have linked discrimination to adverse health behaviors, mental illnesses, heart disease (clinical and preclinical), and metabolic disorders.[5–8] Given the impact of discrimination on physical health, it is plausible that this psychosocial stressor may also increase mortality risk. For instance, studies reported that tobacco use, type 2 diabetes, and cardiovascular disease were each associated with greater experiences of discrimination and increased mortality risk.[5,6,9–11] Among the limited studies that have examined the discrimination-mortality relationship, findings have included null, positive, and inverse associations.[12–14] These inconsistent findings may reflect differences in the study populations (gender, race, and ethnicity) and variations in measurement of discrimination. Studies that have reported associations between discrimination and incident cardiovascular disease, possibly operating through behavioral or stress-mediated pathways, suggest that discrimination might also increase risk of cardiovascular mortality.[9,15] However, the discrimination-cardiovascular mortality relationship remains understudied. Additionally, studies have suggested the relationship between discrimination and health may differ by sociodemographic characteristics; however, this potential effect modification of the discrimination-mortality relationship is not well understood.[16]

Discrimination is the result of negative behaviors or acts toward an individual shaped by unfavorable attitudes held toward the group to which the individual belongs. As a result, discrimination often functions as a manifestation of societal hierarchical power of one group over another.[7,16] For instance, racial and ethnic minoritized populations are more likely to report greater experiences of discrimination, where the relationship on adverse health is well documented, particularly among Black individuals.[7,16] Additionally, experiences of discrimination may affect men and women differently, and therefore, there may be gender differences in the effect of discrimination on health, although findings are mixed.[7] Some studies suggest that the association between discrimination and physical health was stronger among women than men, whereas others reported the association was stronger among men.[7,17–19] These findings suggest that the relationship between discrimination and mortality may differ by race and ethnicity and gender.

Racial and ethnic residential segregation (hereafter, residential segregation), which is a manifestation of racism operating at the societal, institutional, and policy levels, may modify the associations between discrimination and mortality given that residential segregation has been linked to both mortality and experiences of discrimination.[2,20–22] Specifically, Black individuals residing in more racially-mixed neighborhoods were more likely to report experiencing discrimination than those residing in predominantly Black neighborhoods.[23,24] Evidence also suggested that racially minoritized populations residing in high own-group racial clustering were protected from the adverse effects of discrimination on health as a result of having more social support.[25,26] However, the role of residential segregation as a modifier of the association between discrimination and mortality is understudied.

The present study prospectively examined the associations of experiences of discrimination (lifetime, everyday) with all-cause and cardiovascular mortality. Moreover, as the relationship between discrimination and adverse health outcomes may differ by sociodemographic characteristics, this study further investigated whether the discrimination-mortality relationship was modified by race and ethnicity, gender, and residential segregation.

METHODS

The present study included data from the MESA (Multi-Ethnic Study of Atherosclerosis). MESA is a large prospective cohort study investigating risk factors for subclinical atherosclerosis among participants at 6 field sites (Baltimore, Maryland; Chicago, Illinois; Forsyth County, North Carolina; Los Angeles, California; New York, New York; and St. Paul, Minnesota). MESA Participants (N=6814) aged 45 to 84 years without a history of clinical cardiovascular disease were enrolled at exam 1 (baseline) from July 2000 to August 2002. Following exam 1 enrollment, participant data were collected at 5 subsequent follow-up exams (exam 2: 2002–2004; exam 3: 2004–005; exam 4: 2005–2007; exam 5: 2010–2011; and exam 6:

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2016–2018). All participants provided informed consent and Institutional Review Board approval was obtained at each study site. Details on the MESA study design have been previously published.[27] The MESA data are available upon request and details regarding data requests can be found at https://www.mesa-nhlbi.org.

In the present study, participants were excluded if they had a prebaseline cardiovascular disease event (n=9). Additionally, individuals with missing data on any of the discrimination measures (n=228), vital status or follow-up exams (n=13), and baseline study covariates (n=258) were excluded from the analysis. Moreover, because of the small number of cardiovascular-specific deaths among Chinese participants, our analyses were restricted to Black participants, Hispanic/Latino participants, and White participants. The final analytic sample included 1633 Black participants, 1403 Hispanic/Latino participants, and 2473 White participants at baseline.

Discrimination Scales

Participants' experiences of discrimination were captured at baseline using the Lifetime Discrimination Scale (a modified version of the Major Experiences of Discrimination Scale) and the Everyday Discrimination Scale. These scales were adapted from the Detroit Area Study and have demonstrated reliability and validity in diverse populations.[28] The lifetime discrimination scale consisted of 6 items that evaluated whether participants ever experienced unfair treatment in their lifetime. Sample items included, "Unfairly prevented from moving into a neighborhood by a landlord or a realtor," "Unfairly fired or denied a promotion" (Table S1).[28,29] A summary score was created based on the 6 scale item responses where each "no" response was assigned a score of 0 and each "yes" response was assigned a score of 1 (range from 0-6) to represent experiences of lifetime discrimination (Cronbach α=0.61).[5,9]

Everyday discrimination was measured using the Everyday Discrimination 9-item instrument to evaluate day-to-day occurrences of unfair treatment.[29] The everyday discrimination scale captured the frequency of specific experiences of unfair treatment occurrence in a respondent's day-to-day life and included items such as "You are treated with less courtesy than other people," "People act as if they think you are not smart" (Table S1). Response options were on a six-point scale: 1=almost every day; 2=at least once a week; 3=a few times a month; 4=a few times a year; 5=less than once a year; and 6=never. Responses to each item were reverse-coded and averaged across all nine items to create an overall score ranging from 1 to 6, where higher scores indicated a greater frequency of day-to-day discrimination (Cronbach α=0.88).

Racial and Ethnic Residential Segregation

Residential segregation was calculated separately by racial and ethnic group using the Getis-Ord Local *Gi** statistic, based on the census tract of the geocoded addresses of the participants at study baseline. The *Gi** statistic returned a *Z* score for each neighborhood, signifying the extent to which the racial and ethnic composition of the focal tract and neighboring tracts deviated from the mean racial and ethnic composition of larger areal units within which the census tracts are nested (counties composing each MESA site).[30] Higher positive *Z* scores represented greater clustering of a particular racial and ethnic group in that tract compared with the larger geographic area (racial and ethnic overrepresentation), *Z* scores near 0 represent racial and ethnic integration, and lower negative *Z* scores represent racial and ethnic underrepresentation relative to the larger areal unit. Additional details on the use of the *Gi** statistic in the MESA cohort have been previously published.[30,31] The *Gi** statistic better reflects both the contextual and spatial aspects of racial and ethnic segregation than studies that rely on the proportion of a racial and ethnic group in a neighborhood as the proxy for racial and ethnic segregation.[22] Residential segregation was based on each participant's own racial and ethnic group at baseline and categorized into 3 levels: (1) low residential segregation (no own-group racial and ethnic clustering characterized by a *Gi** statistic <0); (2) medium residential segregation (*Gi** statistic, 0–1.96); and (3) high residential segregation (high own-group racial and ethnic clustering characterized by a *Gi** statistic >1.96).

Covariates

Before inclusion in the models, the following covariates were evaluated. Information collected at study baseline included age (continuous), gender (man, woman), self-identified race and ethnicity (Black, White, Hispanic/Latino), family income (collapsed categories: $<20 000, 20 000–49 000, or ≥50 000), highest educational attainment (less than high school diploma/General Equivalency Diploma, high school/some college, college degree or higher), nativity status (US-born, foreign-born), health insurance coverage (yes or no), and marital status (married/living with partner, widowed, divorced or separated, never married). Health behavior information collected at baseline included smoking (never, former, or current), alcohol use (never, former, current), and intentional physical activity (based on the MESA Typical Week Physical Activity Survey) measured continuously.[32] Data on health status and comorbidities included hypertension (diastolic [≥90 mm Hg] or systolic [≥140 mm Hg] blood pressure or combined self-reported history of hypertension with hypertension medication use; yes, no), body mass index (weight[kg]/height[m]^2), and diabetes status (based on the American Diabetes Association fasting criteria and medication review; impaired fasting glucose, untreated diabetes, treated diabetes, or no diabetes).[33]

Duration lived in the neighborhood was measured in years. Neighborhood social cohesion at baseline within census tracts was examined at the individual level using the neighborhood social cohesion score which asked MESA participants about their relationship with neighbors based on 5 statements: "Close-knit neighborhood"; "People willing to help their neighbors"; "People in neighborhood don't get along"; "People in neighborhood can be trusted"; and "People in neighborhood do not share same values." Responses to each of the 5 items (ranged from 1 to 5 on level of agreement with each statement) were summed to create the social cohesion variable (ranged from 1 to 25), where greater values indicated higher social cohesion. Neighborhood poverty was assessed based on baseline residential address linked to the Census 2000 data. Neighborhood poverty was defined as the percentage of residents within a neighborhood that was below the US Census Bureau poverty threshold.

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Mortality

The study outcomes included all-cause and cardiovascular mortality. Follow-up time was defined as time (in days) between baseline visit and mortality event or last contact date, whichever occurred first. In the event of study participant death, MESA interviewers contacted the participant's family member to collect information regarding the date and cause of death. This information was verified by a member of the study team using death certificates. All-cause mortality was determined by city or state records and defined as death from any cause. The MESA study committee utilized a standardized definition to determine cause of cardiovascular death.[34] Cardiovascular mortality was defined as death due to atherosclerotic coronary heart disease, stroke, other atherosclerotic disease, or other cardiovascular disease.

Statistical Analysis

Cox proportional hazards regression was used to examine the associations between discrimination (everyday, lifetime) and mortality, where lifetime discrimination and everyday discrimination were included as predictors in separate models for all-cause and cardiovascular mortality. Covariates were selected if their inclusion in the models resulted in a 10% change in the estimated main effect of discrimination. In model 1, we adjusted for sociodemographic characteristics (age, gender, highest educational attainment, race and ethnicity, family income, and study site). We constructed additional models that adjusted for possible mediators (behavioral and clinical) based on prior work.[16,35] Therefore, we further adjusted for health behaviors (model 1+smoking status and intentional physical activity) in model 2, and clinical characteristics (model 2+body mass index, hypertension status, and fasting glucose/diabetes) in model 3. Models that examined the role of residential segregation (model 4) included additional neighborhood-level confounders or mediators (model 3+neighborhood poverty and neighborhood social cohesion). Residential segregation was slightly correlated with neighborhood poverty (r=0.23), although the correlation was not strong enough to exclude neighborhood poverty from the analysis. Additionally, we adjusted for finer categories of income ($<20 000, 20 000–49 999, 50 000–99 999, ≥100 000) in sensitivity analyses to assess whether the discrimination-mortality relationship differed. Moreover, leveraging the longitudinal data structure, we conducted sensitivity analyses including family income, health behaviors, and clinical characteristics as time-varying covariates.

We further evaluated interactions between discrimination measures and gender, race and ethnicity, highest educational attainment, family income, and residential segregation by including a cross-product term for each potential interaction in the fully adjusted models (model 3 for gender, race and ethnicity, highest educational attainment and family income and model 4 for residential segregation). Proportional hazard assumptions were tested with no evidence of violation. All statistical tests were 2-tailed. Analyses were conducted using SAS version 9.4.

RESULTS

Overall, 44.8% of participants reported any experiences of lifetime discrimination (White, 37.4%; Black, 60.9%; and Hispanic/Latino, 39.2%; Table 1). Persons reporting any experiences of lifetime discrimination were younger, more likely to be men, more college-educated, and current smokers than those who did not report at least one experience of lifetime discrimination. Black participants had the highest body mass index and higher likelihood of having hypertension than other racial and ethnic groups, regardless of their experiences of lifetime discrimination. Among the participants with data on neighborhood environment (n=5006), majority of Black and Hispanic/Latino participants resided in areas of high residential segregated neighborhoods regardless of their experiences of lifetime discrimination. Patterns of demographic characteristics were largely analogous to everyday discrimination (Table S2).

Lifetime Discrimination

In the overall population, each increase in reports of lifetime discrimination was associated with 7% higher all-cause mortality (hazard ratio [HR], 1.07 [95% CI, 1.01–1.13]) and 16% higher cardiovascular mortality (HR, 1.16 [95% CI, 1.05–1.29]; Table 2 model 1). The HR attenuated slightly after adjusting for health behaviors (model 2) and clinical characteristics (model 3), but the associations remained after adjusting for all covariates (model 3: all-cause mortality [HR, 1.06 (95% CI, 1.00–1.11)] and cardiovascular mortality [HR, 1.15 (95% CI, 1.04–1.27)]). Additionally, associations were generally similar in men and women for cardiovascular mortality, although the association with cardiovascular mortality was weaker in women than in men.

Among Black participants, each increase in reports of lifetime discrimination (Table 2) was associated with 8% higher all-cause mortality (HR, 1.08 [95% CI, 1.00–1.16]) and 18% higher cardiovascular mortality (HR, 1.18 [95% CI, 1.02–1.37]). These associations did not change substantially after adjustment for health behaviors and clinical characteristics (model 3: all-cause mortality [HR, 1.07 (95% CI, 1.00–1.16)] and cardiovascular mortality [HR, 1.18; (95% CI, 1.02–1.37)]). Similar HRs were observed for Hispanic/Latino participants but HRs for cardiovascular mortality were weaker compared with Black participants, although not statistically significant (model 3: all-cause mortality [HR, 1.08 (95% CI, 0.96–1.22)] and cardiovascular mortality [HR, 1.07 (95% CI, 0.84–1.37)]). Among White participants, in fully adjusted model (model 3) lifetime discrimination was marginally associated with all-cause mortality and remained associated with cardiovascular mortality although confidence intervals were wide and included the null value (model 3: all-cause mortality [HR, 1.02 (95% CI, 0.92–1.13)] and cardiovascular mortality [HR, 1.14 (95% CI, 0.93–1.39)]). When stratified by highest educational attainment, each increase in reports of lifetime discrimination was associated with higher mortality for individuals who completed

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Table 1. Selected Baseline Participant Characteristics by Race and Ethnicity and Experiences of Lifetime Discrimination in the Multi-Ethnic Study of Atherosclerosis, 2000–2002

	Overall (n=5509)		White (n=2473)		Black (n=1633)		Hispanic/Latino (n=1403)	
	None (55.2%)	Any (44.8%)	None (62.6%)	Any (37.4%)	None (39.1%)	Any (60.9%)	None (60.8%)	Any (39.2%)
	% or mean±SD	% or mean±SD	% or mean±SD	% or mean±SD	% or mean±SD	% or mean±SD	% or mean±SD	% or mean±SD
Age, y, mean (SD)	63.3±10.3	60.1±9.8	63.8±10.2	60.1±9.7	63.5±10.0	60.3±9.8	62.3±10.4	59.9±10.1
Gender								
Men	42.9	51.7	47.1	49.1	31.8	51.2	43.5	57.1
Women	57.1	48.3	52.9	50.9	68.2	48.8	56.5	42.9
Highest educational attainment								
<High school diploma/GED	20.6	12.1	6.3	2.4	15.6	8.2	50.4	35.5
High school/some college	43.7	40.9	43.5	34.0	50.1	44.8	39.4	45.6
College degree or higher	35.7	47.0	50.2	63.6	34.3	47.0	10.2	18.9
Nativity status								
US-born (50 United States or Puerto Rico)	76.4	83.4	93.0	93.7	88.4	91.9	50.7	37.0
Foreign-born	23.6	16.6	7.0	6.3	11.6	8.1	49.3	63.0
Health coverage								
Yes	93.2	91.8	97.8	96.7	96.2	92.2	82.6	82.5
No	6.8	8.2	2.2	3.3	3.8	7.8	17.4	17.5
Marital status								
Married/living with partner	61.0	54.7	69.2	61.4	43.7	45.5	59.2	60.4
Widowed	15.2	10.4	12.3	7.3	21.6	13.9	15.5	9.3
Divorced or separated	15.1	23.6	10.4	19.6	23.3	27.1	17.7	23.6
Never married or prefer not to answer	8.7	11.3	8.1	11.7	11.4	13.5	7.6	6.7
Family income								
$<20 000	22.7	19.5	10.9	10.6	26.9	17.9	41.2	37.1
$20 000–49 999	37.3	37.0	31.2	32.1	43.5	39.5	43.6	40.7
$≥50 000	40.0	43.5	57.9	57.3	29.6	42.6	15.2	22.2
Body mass index,* kg/m^2, mean (SD)	28.6±5.3	29.2±5.6	27.5±4.9	28.1±5.4	30.4±6.0	30.0±5.8	29.1±5.0	29.7±5.2
Smoking status								
Never	47.1	40.2	41.1	36.3	48.2	40.7	57.3	46.0
Former	39.1	41.2	45.5	48.2	36.0	37.6	29.8	36.0
Current	13.8	18.6	13.4	15.5	15.8	21.7	12.9	18.0
Alcohol use								
Never	18.8	11.4	11.2	5.4	21.8	12.9	30.4	18.9
Former	23.0	26.3	17.5	19.9	32.0	31.9	26.1	26.7
Current	58.2	62.3	71.3	74.7	46.2	55.2	43.5	54.4
Hypertension								
Yes	46.2	43.4	40.7	34.4	62.6	54.6	44.1	38.2
No	53.8	56.6	59.3	65.6	37.4	45.4	55.9	61.8
Intentional physical activity, MET-min/wk, mean (SD)	1528.6±2360.5	1760.9±2574.5	1670.9±2413.6	1739.3±2117.3	1617.5±2608.0	1864.4±3049.8	1203.5±2013.8	1610.1±2319.6
Fasting glucose/diabetes status								
Normal	76.6	72.5	83.6	82.6	70.4	68.0	68.7	63.6
Impaired fasting glucose	12.2	14.4	10.7	11.9	13.2	14.6	14.3	18.4
Untreated diabetes	2.4	2.8	1.7	1.4	3.0	3.7	3.1	3.5
Treated diabetes	8.8	10.3	4.0	4.1	13.4	13.7	13.9	14.5

(*Continued*)

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Table 1. Continued

	Overall (n=5509)		White (n=2473)		Black (n=1633)		Hispanic/Latino (n=1403)	
	None (55.2%)	Any (44.8%)	None (62.6%)	Any (37.4%)	None (39.1%)	Any (60.9%)	None (60.8%)	Any (39.2%)
	% or mean±SD	% or mean±SD	% or mean±SD	% or mean±SD	% or mean±SD	% or mean±SD	% or mean±SD	% or mean±SD
Everyday discrimination, mean (SD)†	1.5±0.5	1.9±0.8	1.5±0.5	1.8±0.6	1.6±0.7	2.0±0.8	1.3±0.5	1.7±0.8
Lifetime discrimination, mean (SD)‡	0.0±0.0	1.8±1.0	0.0±0.0	1.5±0.8	0.0±0.0	2.0±1.1	0.0±0.0	1.7±0.9
Neighborhood poverty, %, mean (SD)§	0.2±0.1	0.2±0.1	0.1±0.1	0.1±0.1	0.2±0.1	0.2±0.1	0.2±0.1	0.2±0.1
Neighborhood social cohesion, mean (SD)§‖	17.8±2.9	17.4±3.0	18.2±2.7	17.7±2.8	17.9±2.9	17.6±3.0	17.1±2.9	16.8±3.0
Years lived in neighborhood, mean (SD)§	21.1±14.9	18.4±13.5	21.7±15.5	18.8±13.9	21.7±14.4	18.3±12.8	19.4±14.1	18.0±14.0
Own-group residential segregation (G statistic) §¶								
Low (<0)	30.7	29.3	41.7	46.8	19.7	17.9	18.0	19.7
Medium (0–1.96)	33.1	29.0	45.0	40.5	24.4	24.1	17.0	18.0
High (> 1.96)	36.2	41.7	13.3	12.7	55.9	58.0	65.0	62.3
All-cause mortality	26.8	23.1	28.2	20.2	29.1	26.4	22.6	22.2
Cardiovascular mortality	6.1	5.8	5.7	4.9	6.7	7.0	6.5	5.3

Note: Lifetime discrimination category: none: 0.00–0.00; any: 1.00–6.00. GED indicates general equivalency diploma; and MET, metabolic equivalent of task.

*Body mass index was calculated as weight (kg)/height (m)^2

†Higher scores indicate more everyday experiences of discrimination (range, 1–6)

‡Higher scores indicate more lifetime experiences of discrimination (range, 0–6).

§Restricted to participants with data on neighborhood environment (n=5006).

‖Higher scores indicate higher neighborhood social cohesion (range, 5–25).

¶Residential segregation is measured by using the Gi* statistic. For all race and ethnic groups, low segregation=Gi*<0; medium segregation=Gi* 0–1.96; and high segregation=Gi*>1.96.

high school/some college (model 3: all-cause mortality [HR, 1.10 (95% CI, 1.02–1.19)] and cardiovascular mortality [HR, 1.25 (95% CI, 1.08–1.46)] ; Table S3). Moreover, among those with a family income <$20 000, each increase in reports of lifetime discrimination was associated with 13% higher all-cause mortality (model 3: HR, 1.13 [95% CI, 1.03–1.24]).

When examining the modifying role of residential segregation on the relationship between lifetime discrimination and mortality, the largest HR observed within each racial and ethnic group was among those residing in more racially/ethnically mixed neighborhoods (medium residential segregation), though findings were not statistically significant (Table S4). Moreover, there were no significant interactions between lifetime discrimination and race and ethnicity, gender, highest educational attainment, family income, and residential segregation in relation to all-cause or cardiovascular mortality (all P>0.05).

Everyday Discrimination

Increasing reports of everyday discrimination were associated with higher all-cause mortality in the overall population (model 3: HR, 1.07 [95% CI, 0.98–1.17]), although CIs overlapped with the null. Each increase in reports of everyday discrimination was associated with 21% higher (Table 3) cardiovascular mortality (model 1: HR, 1.21 [95% CI, 1.03–1.42]), and the association remained after adjusting for health behaviors (model 2: HR, 1.20 [95% CI, 1.02–1.42]) and clinical characteristics (model 3: HR, 1.21 [95% CI, 1.03–1.43]). Associations of everyday discrimination with cardiovascular mortality were slightly stronger in men than in women (not statistically significant in women).

When stratified by race and ethnicity, associations between everyday discrimination and all-cause mortality were observed among White and Hispanic/Latino participants but not among Black participants, however in all cases, the confidence intervals were wide and overlapped with the null. Similarly, there was a statistically significant association between everyday discrimination and cardiovascular mortality among White participants (HR, 1.56 [95% CI, 1.17–2.10]), which attenuated but remained after adjusting for health behaviors (model 2: 1.55 [95% CI, 1.16–2.08]) and clinical characteristics (model 3: HR, 1.51 [95% CI, 1.13–2.02). Associations between everyday discrimination and cardiovascular mortality were also observed for Hispanic/Latino participants (HR, 1.27 [95% CI, 0.90–1.80]), but confidence intervals included the null. In contrast, no associations were observed for Black participants. In analyses stratified by highest educational attainment, each increase in reports of everyday discrimination among individuals with less than a high school diploma/General Equivalency Diploma was associated with an increased risk of all-cause (HR, 1.30 [95% CI, 1.08–1.56]) and cardiovascular (HR, 1.59 [95% CI, 1.19–2.13]) mortality in the fully adjusted model, although the interaction between

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Table 2. Adjusted HRs of All-Cause and Cardiovascular Mortality Associated With Experiences of Lifetime Discrimination in the Overall Population and Stratified by Gender and Race and Ethnicity, Multi-Ethnic Study of Atherosclerosis, 2000–2018

	Overall* (n=5509)			Men† (n=2580)			Women‡ (n=2929)		
	HR (95% CI)			HR (95% CI)			HR (95% CI)		
	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3
All-cause mortality	1.07 (1.01–1.13)	1.06 (1.01–1.12)	1.06 (1.00–1.11)	1.07 (1.00–1.14)	1.06 (0.99–1.14)	1.05 (0.98–1.13)	1.07 (0.98–1.16)	1.05 (0.96–1.15)	1.05 (0.96–1.15)
Cardiovascular mortality	1.16 (1.05–1.29)	1.16 (1.05–1.29)	1.15 (1.04–1.27)	1.21 (1.06–1.37)	1.21 (1.06–1.37)	1.19 (1.05–1.35)	1.10 (0.92–1.32)	1.09 (0.91–1.31)	1.09 (0.91–1.31)
	White§ (n=2473)			Black‖ (n=1633)			Hispanic/Latino¶ (n=1403)		
	HR (95% CI)			HR (95% CI)			HR (95% CI)		
	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3
All-cause mortality	1.06 (0.95–1.17)	1.05 (0.94–1.16)	1.02 (0.92–1.13)	1.08 (1.00–1.16)	1.07 (0.99–1.15)	1.07 (1.00–1.16)	1.08 (0.97–1.21)	1.09 (0.97–1.23)	1.08 (0.96–1.22)
Cardiovascular mortality	1.20 (0.98–1.46)	1.20 (0.98–1.46)	1.14 (0.93–1.39)	1.18 (1.02–1.37)	1.18 (1.02–1.36)	1.18 (1.02–1.37)	1.07 (0.84–1.35)	1.10 (0.86–1.39)	1.07 (0.84–1.37)

Note: Per each increase in reports of lifetime discrimination, with values ranging from 0 to 6. Model 1: age, gender, highest educational attainment, race and ethnicity, family income, study site. Model 2: model 1+, smoking status, intentional physical activity. Model 3: model 2+, body mass index, hypertension status, fasting glucose/diabetes status. *P* for interaction for model 3. HR indicates hazard ratio.

Number of events:

*Overall: all-cause mortality n=1387; cardiovascular mortality n=330.

†Men: all-cause mortality n=760; cardiovascular mortality n=186.

‡Women: all-cause mortality n=627; cardiovascular mortality n=144.

§White: all-cause mortality n=624; cardiovascular mortality n=133.

‖Black: all-cause mortality n=448; cardiovascular mortality n=113.

¶Hispanic/Latino: all-cause mortality n=315; cardiovascular mortality n=84.

everyday discrimination and highest level of education was only significant for all-cause mortality (model 3: $P_{interaction}$=0.04; Table S5). In addition, when stratified by family income, all-cause, and cardiovascular mortality HRs were greatest for individuals with a family income <$20 000, although not statistically significant (model 3: all-cause mortality [HR, 1.11 (95% CI, 0.95–1.28)] and cardiovascular mortality [HR, 1.28 (95% CI, 0.97–1.67)]).

Residential segregation did not modify the relationship between experiences of everyday discrimination and mortality (all-cause and cardiovascular; Table S6). There were no significant interactions between everyday discrimination and race and ethnicity, gender, family income, and residential segregation in relation to all-cause and cardiovascular mortality (all *P*>0.05).

Sensitivity Analyses

The results for the discrimination-mortality relationship using finer categories of income in the analyses were the same as the results observed in the analyses using the collapsed categories of income (data not shown). Analyses with time-varying covariates for the associations between discrimination and mortality revealed similar results as the analyses without time-varying covariates (Tables S7 and S8).

DISCUSSION

In this multiethnic cohort study, experiences of lifetime discrimination were associated with increased risk of all-cause and cardiovascular mortality in the overall population. The magnitude of the association was generally similar across gender and race and ethnicity, although in racial and ethnic stratified analyses associations were strongest and statistically significant only in Black participants. Experiences of everyday discrimination were also associated with higher risks of all-cause and cardiovascular mortality in the overall population, in men and women, and among White and Hispanic/Latino participants (although in some cases confidence intervals were wide). Everyday discrimination was not associated with increased all-cause and cardiovascular mortality in Black participants. Notably, associations of discrimination measures with mortality were stronger for cardiovascular than for all-cause mortality.

These study findings contribute to the literature on discrimination and physical health by providing greater understanding of the harmful effects of discrimination on mortality. Among the limited studies that examined the relationship between discrimination and mortality, findings were heterogeneous. For example, a prospective cohort of Black women observed no association between perceived racism (adapted from the everyday and major experiences of discrimination scales) and all-cause and cardiovascular mortality.[12] Another prospective study reported an inverse association between perceived discrimination and all-cause mortality among older Black men and women.[14] Differences in findings between these studies and the present study may be due to differences in study cohort age, gender, and racial and ethnic diversity, as well as differences in how discrimination was measured. However, findings from a Chicago-area study

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Table 3. Adjusted HRs of All-Cause and Cardiovascular Mortality Associated with Experiences of Everyday Discrimination in the Overall Population and Stratified by Gender and Race and Ethnicity, Multi-Ethnic Study of Atherosclerosis, 2000–2018

	Overall* (n=5509)			Men† (n=2580)			Women‡ (n=2929)		
	HR (95% CI)			HR (95% CI)			HR (95% CI)		
	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3
All-cause mortality	1.08 (0.99–1.18)	1.07 (0.98–1.16)	1.07 (0.98–1.17)	1.09 (0.97–1.22)	1.07 (0.96–1.20)	1.07 (0.96–1.20)	1.07 (0.93–1.23)	1.06 (0.92–1.22)	1.07 (0.93–1.24)
Cardiovascular mortality	1.21 (1.03–1.42)	1.20 (1.02–1.42)	1.21 (1.03–1.43)	1.25 (1.01–1.53)	1.23 (1.00–1.52)	1.24 (1.01–1.53)	1.16 (0.90–1.51)	1.16 (0.89–1.51)	1.18 (0.89–1.55)
	White§ (n=2473)			Black‖ (n=1633)			Hispanic/Latino¶ (n=1403)		
	HR (95% CI)			HR (95% CI)			HR (95% CI)		
	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3
All-cause mortality	1.15 (0.98–1.34)	1.14 (0.98–1.33)	1.13 (0.97–1.32)	0.98 (0.85–1.12)	0.95 (0.83–1.09)	0.97 (0.84–1.11)	1.15 (0.97–1.36)	1.17 (0.98–1.40)	1.16 (0.97–1.39)
Cardiovascular mortality	1.56 (1.17–2.10)	1.55 (1.16–2.08)	1.51 (1.13–2.02)	0.99 (0.75–1.29)	0.97 (0.74–1.27)	0.99 (0.76–1.30)	1.24 (0.89–1.72)	1.29 (0.92–1.82)	1.27 (0.90–1.80)

Note: Per each increase in reports of everyday discrimination, with values ranging from 1 to 6. Model 1: age, gender, highest educational attainment, race and ethnicity, family income, study site. Model 2: model 1+, smoking status, intentional physical activity. Model 3: model 2+, body mass index, hypertension status, fasting glucose/diabetes status. *P* for interaction for model 3. HR indicates hazard ratio.

Number of events:

*Overall: all-cause mortality n=1387; cardiovascular mortality n=330.

†Men: all-cause mortality n=760; cardiovascular mortality n=186.

‡Women: all-cause mortality n=627; cardiovascular mortality n=144.

§White: all-cause mortality n=624; cardiovascular mortality n=133.

‖Black: all-cause mortality n=448; cardiovascular mortality n=113.

¶Hispanic/Latino: all-cause mortality n=315; cardiovascular mortality n=84.

among Black and White older adults reported that everyday discrimination was marginally associated with all-cause mortality, similar to findings in the present study.[13]

In the present study, we observed largely similar patterns in associations between discrimination measures and increased mortality (all-cause and cardiovascular), where for both measures the strongest association was observed for cardiovascular mortality. The findings are also consistent with prior studies linking greater experiences of psychosocial stressors to preclinical heart disease and incident cardiovascular events, potentially explaining the stronger association observed for cardiovascular mortality in this study.[9,11,16,36,37]

In gender-stratified analyses, no significant interactions were observed between discrimination (lifetime and everyday) and mortality (all-cause and cardiovascular). However, when comparing mortality hazard in gender-stratified analysis, the associations between discrimination (lifetime and everyday) and cardiovascular mortality were greater among men than in women. A potential explanation for the stronger relationship between experiences of discrimination and cardiovascular mortality among men may be due to gender differences in coping style. Evidence suggests that for active coping, women are more likely to utilize social support when exposed to discrimination than men, thereby buffering the adverse effect of stress on health.[15,16] Another explanation is that experiences of discrimination for men and women may be attributed to different causes (eg, gender) and have differing effects on health.[16,19]

In race and ethnicity–stratified analyses, lifetime discrimination was associated with increased cardiovascular mortality across all racial and ethnic groups, but the strongest association was observed among Black participants in all models. The present study findings and prior literature indicate that Black participants (compared to other racial and ethnic groups) were exposed to more major experiences of discrimination over their life course, thus leading to more deleterious health outcomes than their non-Black counterparts.[2,38] Additionally, associations between everyday discrimination and cardiovascular mortality were only statistically significant among White participants. Prior studies that observed associations between discrimination and health among White individuals suggest that these findings may reflect socioeconomic status and ethnic identity. For instance, White individuals that are of low socioeconomic status or belong to ethnic groups that experience historical prejudice (eg, Jewish, Polish) are more likely to encounter experiences of discrimination than their other White counterparts.[39,40] Additionally, we observed an association between everyday discrimination and cardiovascular mortality among Hispanic/Latino participants. Prior studies reported that Hispanic/Latino individuals encounter routine experiences of unfair treatment attributed to their country of birth, immigration status, and language, which may explain the association observed in the present study, where the Hispanic/Latino population was mostly foreign-born.[41] Interestingly, despite higher reporting of discrimination among Black participants, everyday

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discrimination was not associated with mortality in this population. However, these findings are consistent with results from previous studies that did not observe an association between discrimination and mortality.[12,14] These unexpected findings among Black individuals might be due to utilization of effective strategies (eg, social support) to cope with day-to-day experiences of discrimination and adaptation resulting from greater exposure to unfair treatment throughout the life course compared with other racial and ethnic groups, thereby buffering their mortality risk.[7,16]

In analyses stratified by educational attainment, experiences of lifetime discrimination were associated with higher all-cause and cardiovascular mortality across all educational attainment groups, with the strongest association among individuals that completed high school/some college. However, the strongest association between everyday discrimination and mortality (all-cause and cardiovascular) was observed among those who had less than a high school diploma/General Equivalency Diploma. Given that socioeconomic position is often determined by educational attainment, individuals with less education tend to encounter more barriers to health-promoting resources and experience more social exclusion compared to individuals who are more educated.[42] These discriminatory experiences are most notable in the workforce where individuals with less education are more likely to receive lower salaries and work in jobs that often do not provide health insurance coverage.[42,43] Prior studies suggest that lower socioeconomic position contributes to increased risk for chronic disease and engaging in maladaptive behaviors (eg, tobacco use), which are all associated with elevated mortality risk.[42,44] When examining family income, we observed associations between more experiences of discrimination (lifetime and everyday) and increased all-cause mortality, where the strength of the association increased as family income decreased (ie, the strongest association was in the <$20 000 income group). Similar to findings for low educational attainment, studies on income reported that individuals with low family income experienced income-related social exclusion and unfair treatment, thereby affecting health via engaging in health-risk behaviors and experiencing chronic stress, which has direct physiological consequences that contribute to higher mortality.[2,16,45,46] Another potential explanation is that individuals with low family income may be more likely to experience barriers to health-promoting resources as a result of residing in neighborhoods with harmful conditions that compromise health and increase mortality risk (eg, pollution, crime).[47]

Residential segregation, a spatialized form of institutional racism resulting from historical federal policies and government support for private policies (ie, discriminatory zoning, redlining) has been linked to health status and experiencing psychosocial stressors.[2,7,22,23,26] Specifically, high neighborhood clustering of Black residents due to residential segregation in the United States has been linked to poor health outcomes, largely owing to community disinvestment.[2,22] However, residing in neighborhoods with a higher density of persons belonging to one's own racial and ethnic group may also be protective against the effect of discrimination on health among racially minoritized populations.[23–26] In the present study, racial and ethnic residential segregation did not modify the relationship between discrimination (lifetime and everyday) and all-cause mortality, similar to previous studies for hypertension and obesity.[8,11] However, the highest all-cause mortality hazard was observed within each racial and ethnic group for participants residing in a more racially and ethnically–mixed neighborhood for experiences of lifetime discrimination. A potential reason for the lack of statistical significance could be due to low statistical power, therefore, results must be interpreted cautiously.

Although our study provides insight into the relationship between discrimination and mortality, several limitations must be noted. Discrimination was only captured at baseline and not at each follow-up visit, limiting the ability to evaluate potential changes in experiences of discrimination on mortality risk over the follow-up period. Due to small sample size, we were unable to evaluate the impact of specific discrimination attributions (eg, racial and ethnic discrimination, gender discrimination) on mortality, and therefore, unable to accurately assess whether the relationship differed by the type of discrimination experienced.[7] Additionally, although capturing discrimination experiences is complex and includes several domains, we did not explore these domains in our analyses.[7] Furthermore, the sample size in our race and ethnicity–stratified analyses may have been too small to detect interactions. The Cronbach α for the lifetime discrimination scale was relatively low, potentially resulting in lower internal consistency. Although there is evidence to suggest that the overall discrimination scales effectively capture discrimination across racial and ethnic groups, discrimination scale items may perform slightly different for individuals who are not US-born, which may explain the lower prevalence of discrimination observed among the mostly foreign-born Hispanic/Latino participants in this study compared to findings from other studies.[48,49] The present study only collected binary categories for gender (male, female), which hindered the ability to assess the impact of experiences of discrimination on mortality among individuals of other gender identities. Given the small number of cardiovascular mortality events, we were unable to examine whether residential segregation modified the relationship between discrimination and cardiovascular mortality. In addition, the present study did not account for competing risks of death, potentially resulting in biased mortality estimates.[50] Finally, there was not sufficient power to assess the relationship between

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discrimination and cardiovascular mortality among Chinese participants, hindering the ability to evaluate the relationship among this group.

Despite these limitations, the present study has several notable strengths, including using data from a large racially and ethnically diverse prospective cohort representing various US geographical areas. Additionally, the longitudinal nature of the data allowed for examination of temporality in the associations between discrimination and mortality. Experiences of discrimination were captured using 2 scales that measured different dimensions of unfair treatment, which allowed for evaluation of the effects of experiences of everyday and lifetime discrimination on mortality. Moreover, the use of objective measures of diabetes status, body mass index, and residential segregation reduced misclassification bias.

CONCLUSIONS

The present study findings suggest that experiences of discrimination are associated with all-cause and cardiovascular mortality. These results highlight the need for policies and public health interventions aimed at eliminating discrimination and mitigating its adverse effects on health. Future studies exploring the intersectionality of race and ethnicity and gender are needed among a larger and more diverse cohort to better understand population differences in the relationships between frequency and type of discrimination and mortality.

ARTICLE INFORMATION

Received October 11, 2022; accepted February 2, 2023.

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Acknowledgments

The authors thank the other investigators, the staff, and the participants of the MESA (Multi-Ethnic Study of Atherosclerosis) for their valuable contributions. A full list of participating MESA investigators and institutions can be found at http://www.mesa-nhlbi.org.

Sources of Funding

The MESA (Multi-Ethnic Study of Atherosclerosis) study was supported by contracts HHSN268201500003I, N01-HC-95159, N01-HC-95160, N01-HC-95161, N01-HC-95162, N01-HC-95163, N01-HC-95164, N01-HC-95165, N01-HC-95166, N01-HC-95167, N01-HC-95168, and N01-HC-95169 from the National Heart, Lung, and Blood Institute, and by grants UL1-TR-000040, UL1-TR-001079, and UL1-TR-001420 from the National Center for Advancing Translational Sciences. The MESA Neighborhood Study was supported by grant R01-HL071759. Additionally, the work was also supported by the National Institutes of Health Intramural Research Program of the National Cancer Institute and the National Institute on Minority Health and Health Disparities.

Disclosures

The information presented by the authors is their own and this material should not be interpreted as representing the official viewpoint of the U.S. Department of Health and Human Services, the National Institutes of Health, the National Institute on Minority Health and Health Disparities, or the National Cancer Institute. The authors report no conflicts.

Code Availability

The analytical methods for this study are available from the corresponding author upon appropriate request.

Supplemental Material

Tables S1–S8

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